

AI's Full Response

AI-Based Approach for Fraud Detection

To solve this fraud detection problem, the following AI-based approach is recommended:

1. Data Preprocessing

- Handle missing values and outliers
- Normalize numerical features like transaction amount
- Encode categorical features (location, device type)

2. Handling Imbalanced Data

- Use techniques like SMOTE (Synthetic Minority Over-sampling Technique)
- Alternatively use undersampling or class-weighted models

3. Feature Engineering

- Extract time-based features (hour, day, weekend)
- Create behavioral features from transaction history
- Detect unusual spending patterns

4. Model Selection

- Start with Logistic Regression (baseline)

- Use tree-based models like Random Forest and Gradient Boosting (XGBoost, LightGBM)
- These models perform well on tabular and imbalanced data

5. Evaluation Metrics

- Accuracy is not suitable due to imbalance
- Use Precision, Recall, F1-score
- ROC-AUC and PR-AUC are important

6. Model Tuning

- Use cross-validation
- Hyperparameter tuning with Grid Search or Random Search

7. Deployment

- Build API for real-time prediction
- Monitor model performance and retrain periodically

8. End-to-End Pipeline

- Data collection → preprocessing → feature engineering → training → evaluation → deployment → monitoring

The most suitable models are Gradient Boosting models (like XGBoost or LightGBM) due to their strong performance on structured data and ability to handle imbalance.

System Architecture Design

